

Radial variation in pernambuco (*Caesalpinia echinata*) wood properties

Laurence R. Schimleck^{1,*}, Jorge Luis Monteiro de Matos² and Charles Espey³

¹Warnell School of Forestry and Natural Resources, The University of Georgia, Athens GA 30602, U.S.A. Current address: Department of Wood Science and Engineering, Oregon State University, Corvallis OR, 97331, U.S.A.

²Department of Forestry and Wood Sciences, Federal University of Parana, Curitiba, PR, Brazil.

³Charles Espey Ltd., Port Townsend, WA, U.S.A.

*Corresponding author; e-mail: laurence.schimleck@oregonstate.edu

ABSTRACT

Caesalpinia echinata Lam. (pernambuco or pau-brasil) is recognized as the premier raw material for manufacturing stringed instrument bows. Several studies have identified properties considered important in determining the suitability of pernambuco wood for bow manufacture including density, modulus of elasticity (MOE), and, possibly, microfibril angle (MFA). No research has been conducted on how these properties vary within individual trees; however, an understanding of how pernambuco wood properties vary within trees is important as it may assist in the identification of trees or provenances most suited for the establishment of plantations, aid in developing an understanding of management practices on wood property variation for plantation-grown pernambuco and also facilitate the identification of regions within trees that possess optimal properties for bow manufacture. Radial variation in density, MFA and MOE was examined using SilviScan for three radial strips representing differing levels of wood quality in terms of perceived suitability for making high-quality bows. The lowest quality sample showed considerable radial variation compared to the higher quality samples for all properties and it also had the lowest average density. It was not possible to identify a strong pith to bark trend for any of the wood properties examined.

Keywords: *Caesalpinia echinata*, density, microfibril angle, modulus of elasticity, pernambuco, SilviScan, wood properties.

INTRODUCTION

Caesalpinia echinata Lam. (pernambuco or pau-brasil), a legume species found in the Atlantic forest of Brazil, is recognized as the premier wood for making stringed instrument bows (Brown 1978; Rymer 2004). Several studies (Matsunaga *et al.* 1996; Angyalossy *et al.* 2005; Alves *et al.* 2008; Schimleck *et al.* 2009) have quantified properties considered important in determining the suitability of pernambuco wood for bow manufacture. However, none have examined how wood properties vary radially within a tree. Considering the scarcity of remaining trees and the difficulty in obtain-

ing any samples for scientific study this is understandable and potentially impossible to achieve on a large scale. An option available for obtaining pernambuco samples for the study of radial variation is to use off-cuts obtained from boards utilized by bow makers for manufacturing bows. Unfortunately, the height in the tree at which boards are collected is unknown but at least they can give an indication of how pernambuco wood properties vary radially in the absence of better characterized samples.

SilviScan (Evans 1998, 1999, 2006) provides a rapid, high resolution option for examining radial variation of a number of wood properties including density, stiffness (modulus of elasticity, MOE) and microfibril angle (MFA). Density is considered a critical property in determining bow quality. Alves *et al.* (2008) reports that only sticks with densities greater than 1,000 kg/m³ are suitable for quality bows; however, in a practical sense, there is an upper density limit with bows made from very high-density wood being excessively thin (to meet the desired weight requirements), making the bow less stable and more susceptible to breakage (Alves *et al.* 2008). In addition to density, high stiffness and strength (modulus of rupture, MOR) are also important (Wegst 2006; Alves *et al.* 2008; Schimleck *et al.* 2009). The role of MFA in determining the suitability of pernambuco wood for the manufacture of high-quality bows is less clear. MFA is an important determinant of wood stiffness (Cave & Walker 1994; Evans & Ilic 2001), hence it could be expected to be important in determining bow quality; however, Schimleck *et al.* (2009) found that it had little influence owing to its very narrow range (6.1 to 14.6 deg.).

The future availability of pernambuco for bow manufacture is uncertain and pernambuco has been CITES - Convention on International Trade in Endangered Species of Wild Fauna and Flora - Appendix II listed (<http://www.cites.org>). Interest exists in using plantation-grown pernambuco as an alternative and a group of bow makers have formed the International Pernambuco Conservation Initiative (IPCI) which aims to conserve the species by establishing plantations for future harvest (Rymer 2004). For such efforts to be successful an understanding of wood property variation in natural populations (provenance variation) and within trees is essential.

Hence the aim of this study was to use SilviScan to examine radial variation of some wood properties considered important in determining the quality of pernambuco wood for bow manufacture and to provide preliminary data on how these properties vary with increasing distance from the pith.

MATERIALS AND METHODS

Sample origin

Three radial strips representing differing levels of wood quality in terms of perceived suitability for making high-quality bows were selected from a group of 13 samples provided by Charles Espey. The samples were cut from the ends of pernambuco boards which had been purchased between 1981 and 1988 and represented a range from very poor (sample 1) to high (samples 2 and 3) suitability for the manufacture of high-quality violin bows.

It should be noted that, owing to the way pernambuco is sawmilled and later sold, it is impossible for the person purchasing boards, *i.e.* the bowmaker, to know specific

information regarding the origin of the trees that the samples came from or the height in the tree at which the boards were cut. The following descriptions were provided by Charles Espey. Information relating to likely origin of the samples is based on his 30+ years experience as a bow maker.

A description of each sample follows:

- 1) Obtained from a board milled from a half log purchased in Seattle in 1981. This wood is not suitable for high-quality bows due to its low strength, low density and interlocked grain.
- 2) Cut from a board rated as excellent for the manufacture of high-quality bows and purchased in 1988.
- 3) Cut from a board purchased in 1984 and rated as excellent for the manufacture of high-quality bows.

Sample dimensions varied but were approximately 10 mm longitudinally and 10 mm tangentially. Radial length varied, sample 1 was 220 mm long, while samples 2 and 3 were both 78 mm in length.

SilviScan analysis

The samples were sent to FPInnovations Paprican (Vancouver, Canada) for SilviScan analysis. Radial strips (2 mm tangentially, 7 mm longitudinally, radial length varied and matched the lengths described for samples 1 to 3) were cut from the samples using a twin blade saw for analysis.

The following three wood properties were measured:

- Air-dry density (kg/m^3) in 25 μ steps using X-ray densitometry;
- MFA (deg.) over 5 mm intervals using scanning X-ray diffractometry (Evans 1998, 1999);
- Wood stiffness (GPa) estimated at the same resolution as MFA by combining X-ray densitometry and X-ray diffraction data (Evans 2006).

The SilviScan measurements were made in a controlled environment of 40 % relative humidity and a temperature of 20 °C.

RESULTS

Radial variation for each of the wood properties measured by SilviScan for the 3 samples are shown in Figures 1 (density), 2 (MFA) and 3 (MOE). Densities for sample 1 averaged 979 kg/m^3 and fluctuated considerably from pith to bark with a minimum of 873 kg/m^3 and maximum of 1054 kg/m^3 . Samples 2 and 3, in comparison, had higher average densities (1116 and 1147 kg/m^3 , respectively) and demonstrated less radial variation. MFA also fluctuated from pith to bark for sample 1, but the difference between minimum (7 deg.) and maximum (11 deg.) MFA's was small. MFA was very consistent for sample 2 with an average of 8.9 deg., while sample 3 showed more variation with a maximum MFA of 16 deg. near the pith and a minimum MFA of 9.4 deg. close to the bark. MOE tended to decrease from pith to bark for sample 1, while it increased for samples 2 and 3. Sample 3 had the highest average MOE (21.7 GPa) while average

MOE's were similar for samples 1 and 2 (19.8 GPa). It was not possible to identify a strong pith to bark trend for any of the examined wood properties.

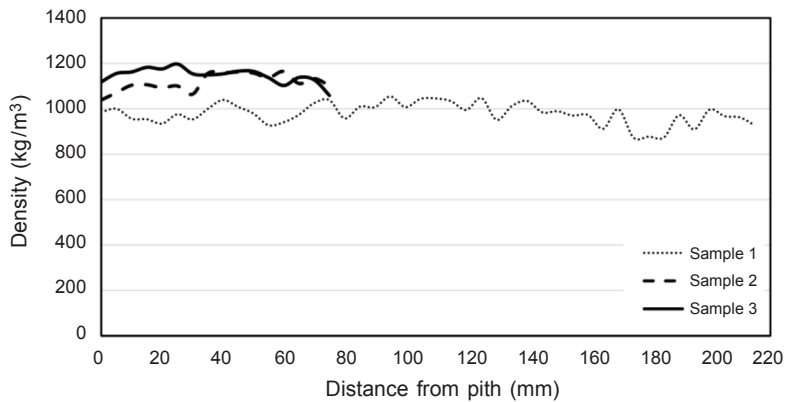


Figure 1. Density radial variation for three pernambuco samples.

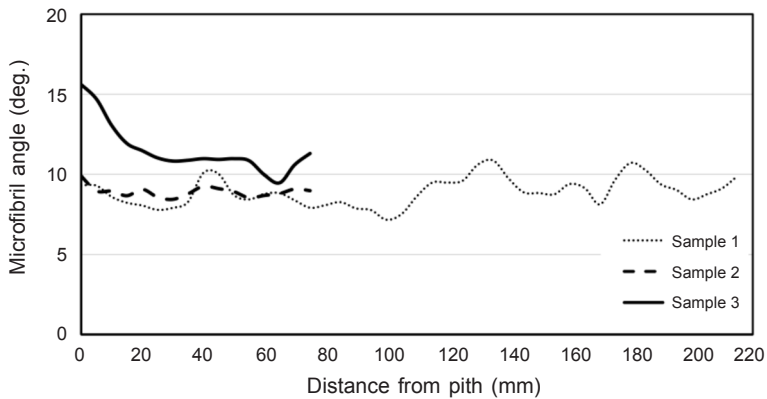


Figure 2. Microfibril angle (MFA) radial variation for three pernambuco samples.

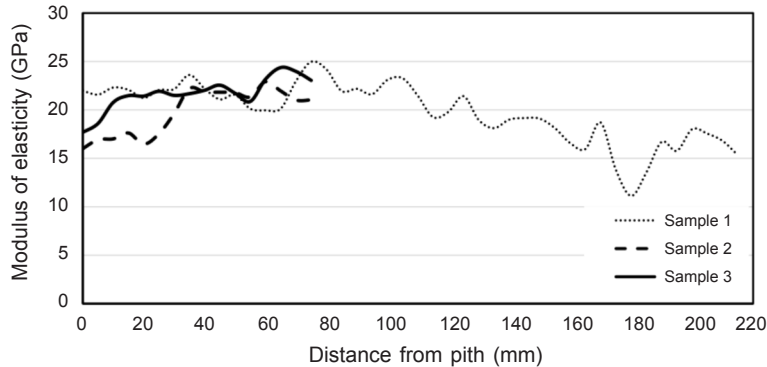


Figure 3. Modulus of elasticity (MOE) radial variation for three pernambuco samples.

DISCUSSION

This study is the first to report patterns of radial variation for wood properties in pernambuco. Knowledge of typical patterns of variation in important wood properties could aid in the identification of trees or provenances with the greatest chance of providing bow quality wood from plantation-grown trees in the future. In addition, such information could greatly assist in understanding the effect of management practices on wood property variation within trees and may also indicate an approximate age (assuming it can be identified), or perhaps distance from pith, at which the production of bow quality wood commences.

Knowledge of within-tree variation of important wood properties could also aid utilization of pernambuco as regions within trees possessing the combination of properties required for the manufacture of high-quality bows could be identified. Densities of at least 1000 kg/m³ are considered to be necessary for the manufacture of quality bows (Alves *et al.* 2008) so it could be expected that sample 1 would be ranked low in terms of quality. What is also of interest, and would not have been apparent without examining radial variation, was the considerable fluctuations that occurred from pith to bark for density in sample 1 compared to sample 2 and the larger difference between maximum and minimum densities (182 kg/m³ for sample 1, 126 kg/m³ for sample 2). Hence, it appears that the desirable radial trend for bows is low pith to bark variability, therefore if plantations were to be established, it would be important to manage growth to minimize variation.

Differences in radial trends arising from the growth environment of a tree is of particular interest considering efforts by the IPCI in establishing pernambuco plantations for future harvest (Rymer 2004). We assume that a desirable outcome for trees from these plantations would be radial trends in properties similar to those for native forest pernambuco which would require careful attention to the silvicultural system used. Interestingly, wood from pernambuco plantations (not established by the ICPI) aged 25 and 30 years was recently examined by Schimleck *et al.* (2013) and compared with native forest pernambuco of varying qualities. While not of the standard of the highest quality wood, plantation grown pernambuco provided promising results.

The radial variation in density, MFA and MOE for three pernambuco samples was examined using SilviScan. For the three properties the identification of a strong pith to bark trend was not possible. The high density and stiffness and low MFA observed in this study were in agreement with earlier studies (Alves *et al.* 2008; Schimleck *et al.* 2009).

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